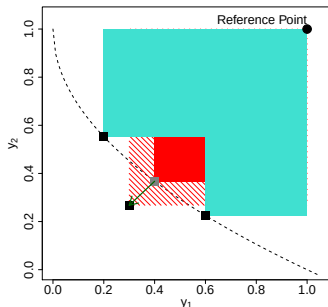
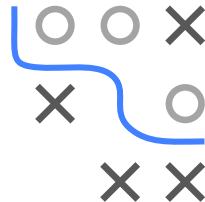


Optimization in Machine Learning

Bayesian Optimization

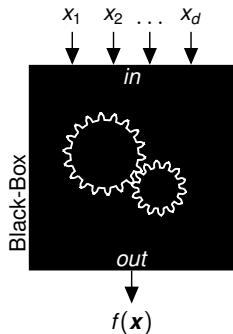
Multicriteria Bayesian Optimization



Learning goals

- Multicriteria Optimization
- Taxonomy
- ParEGO, SMS-EGO, EHI

MULTICRITERIA BAYESIAN OPTIMIZATION



$$f : \mathcal{S} \rightarrow \mathbb{R}^m$$

$$\min_{\mathbf{x} \in \mathcal{S}} f(\mathbf{x}) = (f_1(\mathbf{x}), \dots, f_m(\mathbf{x}))$$

- A configuration $\mathbf{x}$ **dominates** ($\prec$) $\tilde{\mathbf{x}}$ if

$$\forall i \in \{1, \dots, m\} : f_i(\mathbf{x}) \leq f_i(\tilde{\mathbf{x}})$$

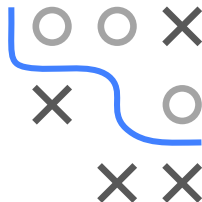
and $\exists j \in \{1, \dots, m\} : f_j(\mathbf{x}) < f_j(\tilde{\mathbf{x}})$

- Set of non-dominated solutions:

$$\mathcal{P} := \{\mathbf{x} \in \mathcal{S} \mid \nexists \tilde{\mathbf{x}} \in \mathcal{S} : \tilde{\mathbf{x}} \prec \mathbf{x}\}$$

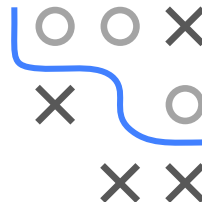
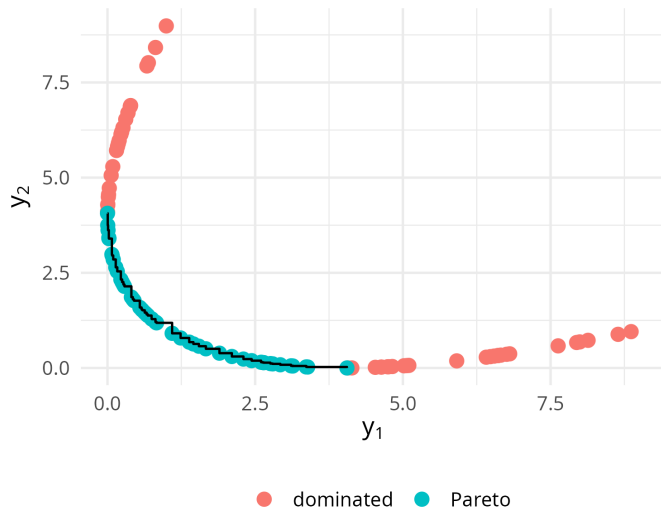
- Pareto set $\mathcal{P}$, Pareto front $\mathcal{F} = f(\mathcal{P})$

- Goal: Find $\hat{\mathcal{P}}$ of non-dominated points that estimates the true Pareto set $\mathcal{P}$



MULTICRITERIA BAYESIAN OPTIMIZATION

Example Pareto front:



MULTICRITERIA BAYESIAN OPTIMIZATION

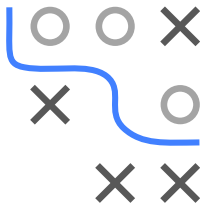
The most popular quality indicator is the hypervolume indicator (also called dominated hypervolume or $\mathcal{S}$ -metric).

The hypervolume, HV, of an approximation of the Pareto front $\hat{\mathcal{F}} = f(\hat{\mathcal{P}})$ can be defined as the combined volume of the dominated hypercubes domHC_r of all solution points $\mathbf{x} \in \hat{\mathcal{P}}$ regarding a reference point $\mathbf{r}$, i.e.,

$$\text{HV}_r(\hat{\mathcal{P}}) := \mu \left(\bigcup_{\mathbf{x} \in \hat{\mathcal{P}}} \text{domHC}_r(\mathbf{x}) \right)$$

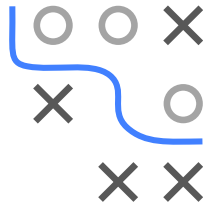
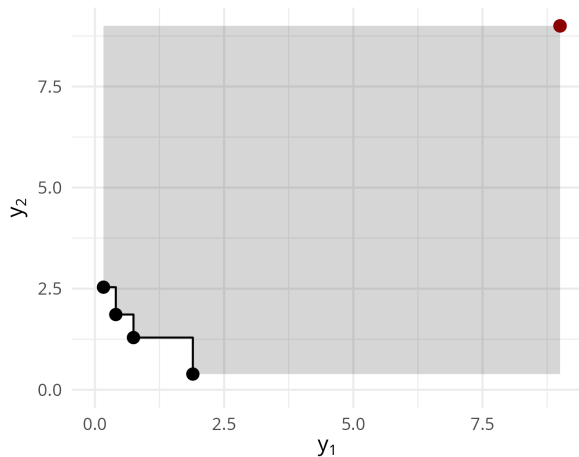
where μ is the Lebesgue measure and the dominated hypercube is given as:

$$\text{domHC}_r(\mathbf{x}) := \{ \mathbf{u} \in \mathbb{R}^m \mid f_i(\mathbf{x}) \leq u_i \leq r_i \forall i \in \{1, \dots, m\} \}$$



MULTICRITERIA BAYESIAN OPTIMIZATION

Hypervolume example:



Reference point $\mathbf{r}$ in red, estimated Pareto front $\hat{\mathcal{F}}$ in black,
corresponding $HV_{\mathbf{r}}(\hat{\mathcal{P}})$ is given by the grey area

```
graph TD
    subgraph Initial_design [Initial design]
        A1[Arbitrary DOE  
(e. g. random,  
Sobol, grid)]
        A2[Latin Hypercube  
Sampling]
    end

    subgraph Model_fitting [Model fitting]
        B1[Individual models for  
each objective]
        B2[Scalarize objectives]
        B3[Model of scalarization]
        A1 --> B1
        A2 --> B3
        B2 --> B3
    end

    subgraph Candidate_generation [Candidate generation]
        C1[Multi-obj.  
optimization of  
infill criterion  
on each model]
        C2[Multiple single-  
obj. optim. of  
(aggregating)  
infill criterion]
        C3[Single-obj.  
optimization of  
(aggregating)  
infill criterion]
        B1 --> C1
        B2 --> C2
        B3 --> C3
    end

    subgraph Candidate_selection [Candidate selection]
        D1[Select design  
point(s) with  
regard to  
other criterion]
        D2[Select design  
point(s) with  
regard to  
infill criterion]
        C1 --> D1
        C2 --> D1
        C3 --> D2
    end

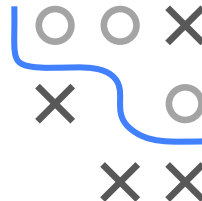
    subgraph Stopping_decision [Stopping decision]
        E1[Decide  
termination  
based on  
approximation]
        E2[Decide  
termination  
based on  
total budget]
        D1 --> E1
        D2 --> E2
    end

    E1 --> End(( ))
    E2 --> End
    E2 -- Update design --> B3
```

The flowchart illustrates the multi-objective optimization process, organized into five horizontal stages:

- Initial design:** Starts with two options: "Arbitrary DOE (e. g. random, Sobol, grid)" and "Latin Hypercube Sampling".
- Model fitting:** The "Arbitrary DOE" path leads to "Individual models for each objective". The "Latin Hypercube Sampling" path leads to "Model of scalarization". "Scalarize objectives" also feeds into "Model of scalarization".
- Candidate generation:** "Individual models for each objective" leads to "Multi-obj. optimization of infill criterion on each model". "Scalarize objectives" leads to "Multiple single-obj. optim. of (aggregating) infill criterion". "Model of scalarization" leads to "Single-obj. optimization of (aggregating) infill criterion".
- Candidate selection:** "Multi-obj. optimization of infill criterion on each model" and "Multiple single-obj. optim. of (aggregating) infill criterion" both lead to "Select design point(s) with regard to other criterion". "Single-obj. optimization of (aggregating) infill criterion" leads to "Select design point(s) with regard to infill criterion".
- Stopping decision:** "Select design point(s) with regard to other criterion" leads to "Decide termination based on approximation". "Select design point(s) with regard to infill criterion" leads to "Decide termination based on total budget".

Additional flow details: A feedback loop labeled "Update design" connects the "Decide termination based on total budget" box back to the "Model of scalarization" box in the Model fitting stage. The process terminates after either decision box in the Stopping decision stage.



```

graph TD
    subgraph Initial_design [Initial design]
        A[Arbitrary DOE  
(e. g. random,  
Sobol, grid)]
        B[Latin Hypercube Sampling]
    end

    subgraph Model_fitting [Model fitting]
        C[Individual models for each objective]
        D[Scalarize objectives]
        E[Model of scalarization]
    end

    subgraph Candidate_generation [Candidate generation]
        F[Multi-obj. optimization of infill criterion on each model]
        G[Multiple single-obj. optim. of (aggregating) infill criterion]
        H[Single-obj. optimization of (aggregating) infill criterion]
    end

    subgraph Candidate_selection [Candidate selection]
        I[Select design point(s) with regard to other criterion]
        J[Select design point(s) with regard to infill criterion]
    end

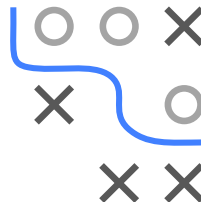
    subgraph Stopping_decision [Stopping decision]
        K[Decide termination based on approximation]
        L[Decide termination based on total budget]
    end

    A --> C
    A --> D
    B --> E
    C --> F
    D --> G
    D --> E
    E --> H
    F --> I
    G --> I
    G --> J
    H --> J
    I --> K
    J --> L
    K --> L
    L --> L
    L --> Update[Update design]
    Update --> B
  
```

The flowchart illustrates the multi-objective optimization process, organized into five main stages:

- Initial design:**
 - Arbitrary DOE (e. g. random, Sobol, grid)
 - Latin Hypercube Sampling
- Model fitting:**
 - Individual models for each objective
 - Scalarize objectives
 - Model of scalarization
- Candidate generation:**
 - Multi-obj. optimization of infill criterion on each model
 - Multiple single-obj. optim. of (aggregating) infill criterion
 - Single-obj. optimization of (aggregating) infill criterion
- Candidate selection:**
 - Select design point(s) with regard to other criterion
 - Select design point(s) with regard to infill criterion
- Stopping decision:**
 - Decide termination based on approximation
 - Decide termination based on total budget

The process includes feedback loops: from the 'Decide termination based on total budget' step, an 'Update design' step leads back to the 'Latin Hypercube Sampling' step in the Initial design stage. Additionally, a dashed arrow indicates a feedback loop from the 'Single-obj. optimization of (aggregating) infill criterion' step back to the 'Scalarize objectives' step.



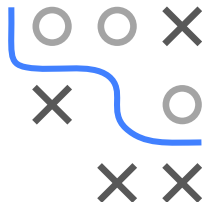
PAREGO

- 1 Scalarize standardized objectives using the augmented Tchebycheff norm

$$\max_{i \in \{1, \dots, m\}} w_i f_i(\mathbf{x}) + \rho \sum_{i=1}^m w_i f_i(\mathbf{x})$$

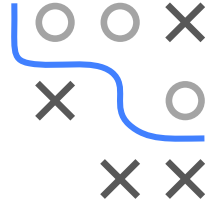
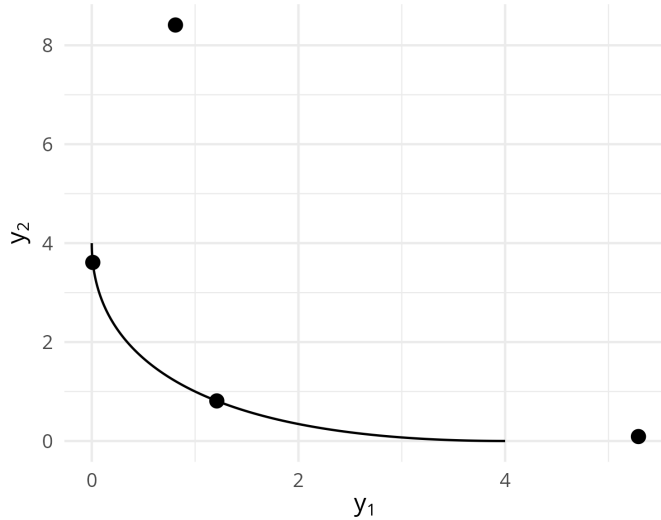
with weight vector $\mathbf{w}$ drawn uniformly from the set of evenly distributed weight vectors $\mathcal{W}$

- 2 Fit SM on the scalarized objective function
- 3 Proceed to use any standard single-objective acquisition function (EI, PI, LCB, ...)



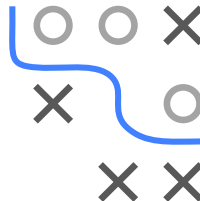
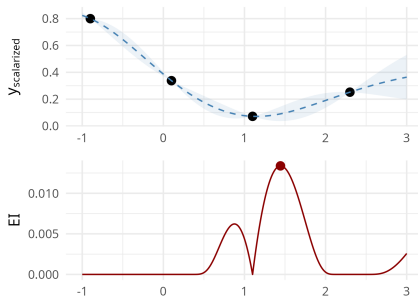
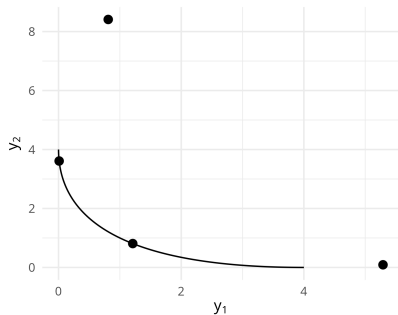
PAREGO

ParEGO Example, initial design and true Pareto front in black ...



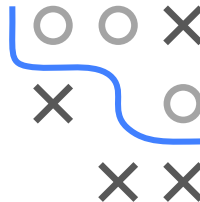
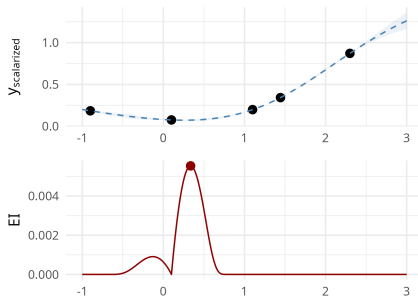
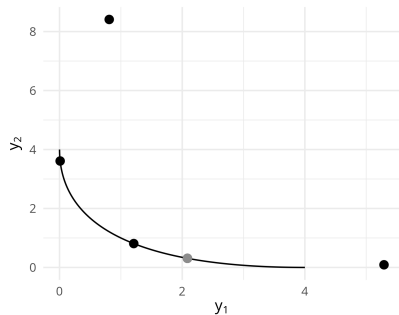
PAREGO

... standardize objectives, obtain scalarized objective via augmented Tchebycheff norm, fit SM and optimize EI ...



PAREGO

... note that the specific scalarization is different at each iteration!



The grey point visualizes the candidate we choose to evaluate in the previous iteration

```

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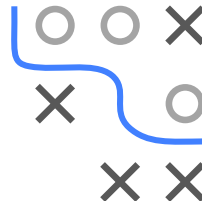
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        K[Decide  
termination  
based on  
approximation]
        L[Decide  
termination  
based on  
total budget]
    end

    Update[Update  
design]

    A --> C
    A --> D
    A --> E
    B --> E
    C --> F
    D --> G
    E --> H
    F --> I
    G --> J
    H --> J
    I --> K
    J --> L
    K --> Update
    L --> Update
    Update --> A
    Update --> B
  
```

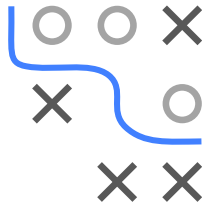
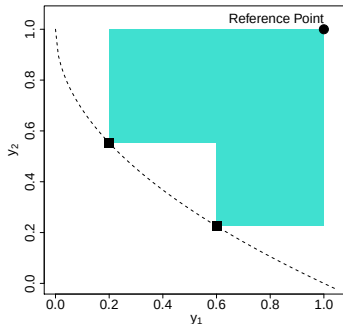


SMS-EGO

Individual models for each objective f_i

Single-objective optimization of aggregating acquisition function:
Calculate contribution of the confidence bound of candidate to the current front approximation

- Calculate LCB for each objective
- Measure contribution with regard to the hypervolume improvement
- For ε -dominated ($\prec_\varepsilon$) solutions, a penalty is added

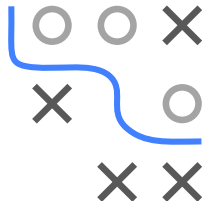
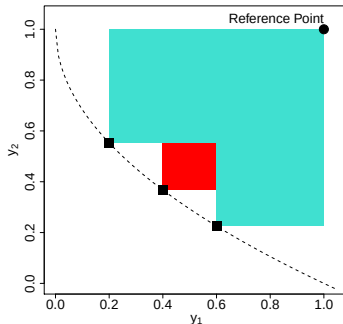


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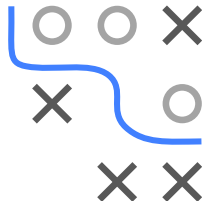
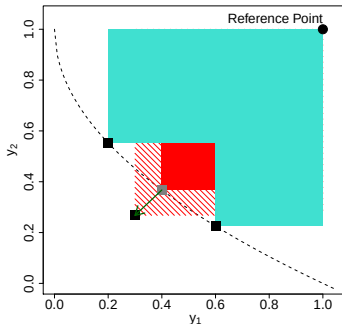


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- Measure contribution with regard to the hypervolume improvement
- For ε -dominated ($\prec_\varepsilon$) solutions, a penalty is added



OUTLOOK

- Many more options exist:
 - Expected Hypervolume Improvement
 - Multi-EGO
 - Entropy based: PESMO, MESMO
 - ...

